| Practical: 4 | Implementation and Time analysis of factorial program using iterative and recursive method |
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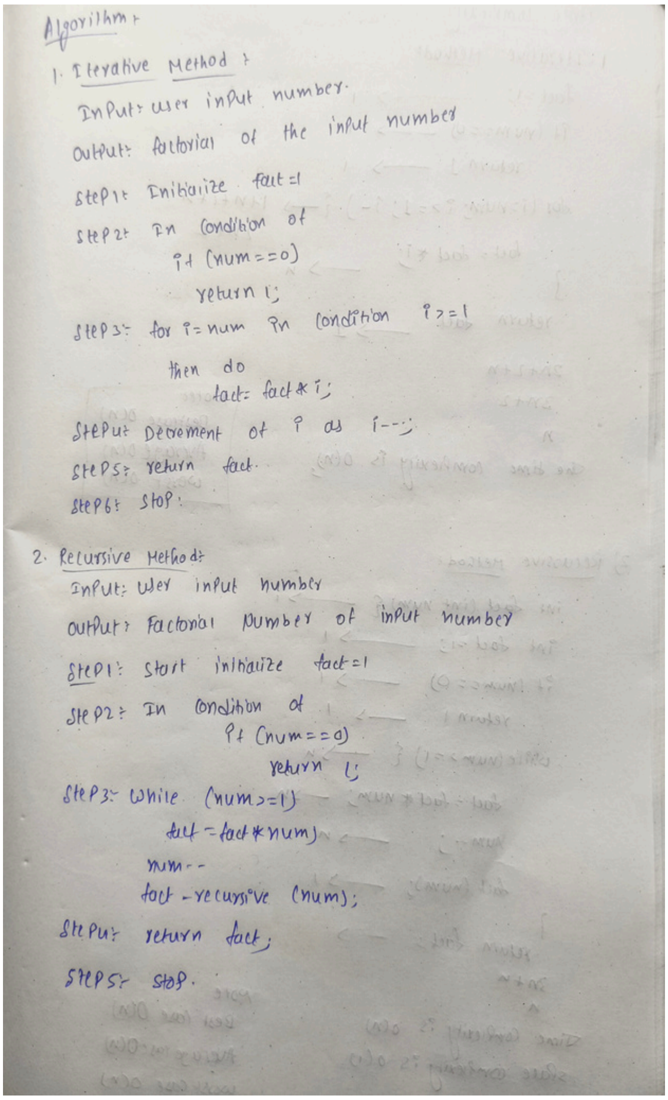
| 18/08/2026 | |

**Aim:** ToImplement and Time analysis of factorial program using iterative and recursive method.

**Algorithm:**


Algorithm :
1. Iterative Method :
Input: user input number.
Output: factorial of the input number
Step1: Initialize fact=1
Step2: In condition of
if (num==0)
return 1;
Step3: for i=num in condition i>=1
then do
fact= fact*i;
Step4: Decrement of i as i--;
Step5: return fact.
Step6: Stop.

2. Recursive Method:
Input: user input number
Output: Factorial Number of input number
Step1: Start initialize fact=1
Step2: In condition of
if (num==0)
return 1;
Step3: while (num>=1)
fact=fact*num;
num--
fact-recursive (num);
Step4: return fact;
Step5: Stop.

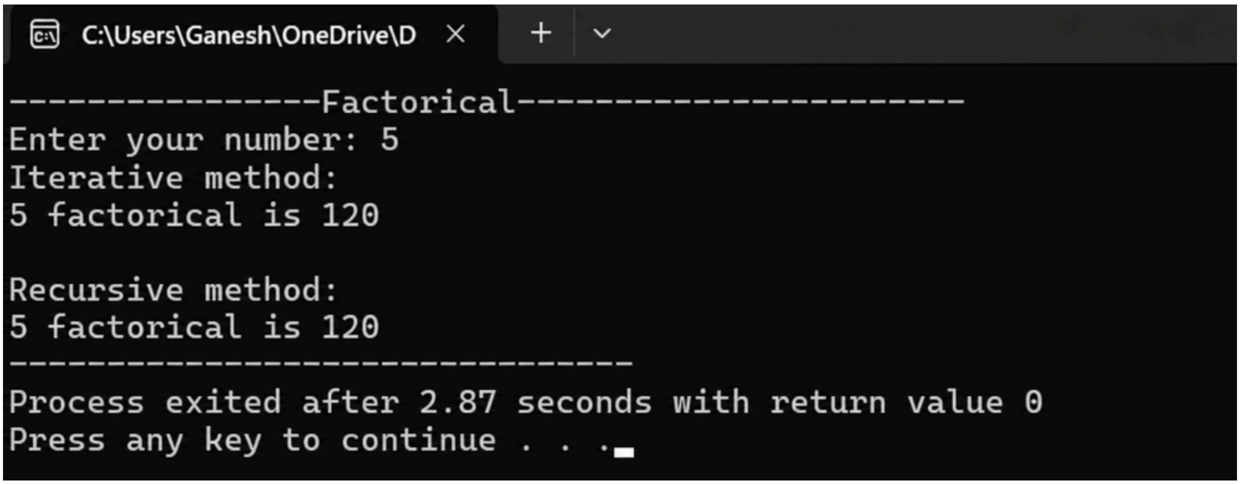
**Program:**

**factorial program using iterative and recursive method**

```
#include<iostream>
using namespace std;
```

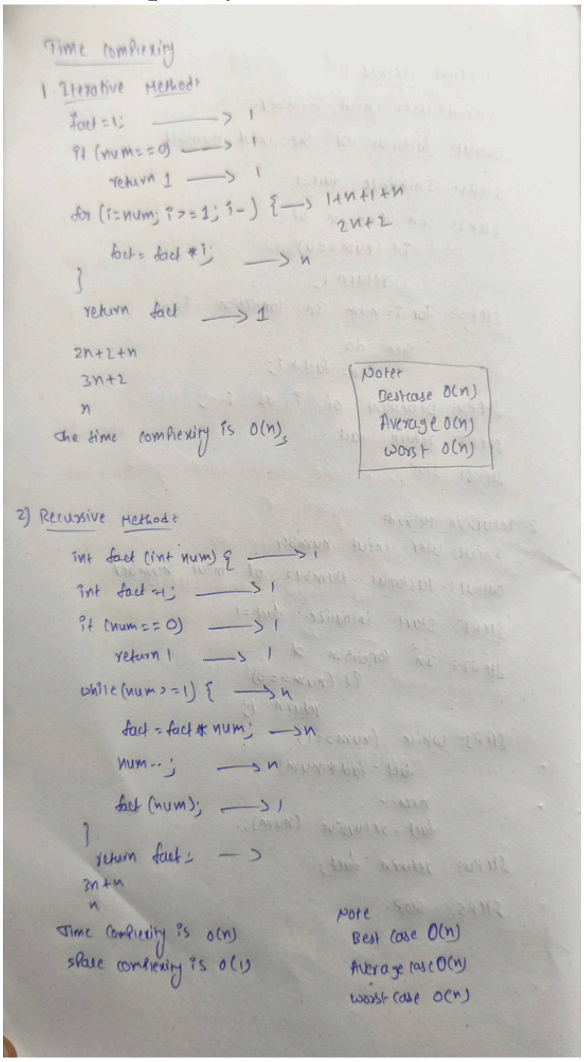
```
intfact_iterative(int num){
        int fact = 1;
        if(num == 0)
                return 1;
        for(int i=num; i>=1; i--){
                fact = fact*i;
        }
        return fact;
}

intfact_recursive(int num){
        int fact = 1;
        if(num == 0)
                return 1;
        while(num >= 1){
                fact = fact*num;
                num--;
                fact_recursive(num);
        }
        return fact;
}

intmain(){
        cout<<"----------------Factorical-----------------------"<<endl;
        int n;
        cout<<"Enter your number: ";
        cin>>n;
        int i = fact_iterative(n);

        cout<<"Iterative method:"<<endl;
        cout<<n<<" factorical is "<<i;
        cout<<endl<<endl;
        int r = fact_recursive(n);
        cout<<"Recursive method:"<<endl;
        cout<<n<<" factorical is "<<r;

        return 0;
}
```

**Output:**



```
C:\Users\Ganesh\OneDrive\D
---------------Factorical----------------------
Enter your number: 5
Iterative method:
5 factorical is 120

Recursive method:
5 factorical is 120
--------------------------------
Process exited after 2.87 seconds with return value 0
Press any key to continue . . .
```

**Time Complexity:**



Time complexity

1. Iterative Method:

fact=1; ⟶ 1
if (num==0) ⟶ 1
return 1 ⟶ 1
for (i=num; i>=1; i--) { ⟶ 1+n+1+n
2n+2
fact= fact *i; ⟶ n
}
return fact ⟶ 1

2n+2+n
3n+2
n
The time complexity is O(n)

Note:
Bestcase O(n)
Average O(n)
Worst O(n)

2) Recursive Method:

int fact (int num) { ⟶ 1
int fact =1; ⟶ 1
if (num==0) ⟶ 1
return 1 ⟶ 1
while (num>=1) { ⟶ n
fact = fact * num; ⟶ n
num--; ⟶ n
fact (num); ⟶ 1
}
return fact; ⟶
3n+n
n
Time complexity is O(n)
space complexity is O(1)

Note
Best case O(n)
Average case O(n)
worst case O(n)

**Conclusion:**

The implementation of this program and the execution is successfully got executed and the time complexity in the Three cases are O(n) in the both Iterative, recursive methods And Space complexity of both methods are O(1).